

FORM VI MATHEMATICS

Time allowed: 2 hours

Exam date: 12th May 2003

Instructions:

- All questions may be attempted.
- All questions are of equal value.
- Part marks are shown in boxes in the right margin.
- All necessary working must be shown.
- Marks may not be awarded for careless or badly arranged work.
- Approved calculators and templates may be used.
- A list of standard integrals is provided at the end of the examination paper.

Collection:

- The writing booklets will be collected in one bundle.
- Start each question in a new writing booklet.
- If you use a second booklet for a question, place it inside the first. Don't staple.
- Write your candidate number on each booklet.

Checklist:

- SGS Writing Booklets required — 8 per boy.
- Candidature 123 boys.

QUESTION ONE (Start a new writing booklet)

- (a) Use your calculator to evaluate each of the following, correct to three decimal places: Marks
- | | |
|------------------|---|
| (i) e^2 | 1 |
| (ii) $\log_e 14$ | 1 |
| (iii) $\sin 1.5$ | 1 |
- (b) Write down the exact value of $\cos \frac{5\pi}{6}$. 2
- (c) Differentiate the following:
- | | |
|------------------------|---|
| (i) $y = 2x^{-1}$ | 1 |
| (ii) $y = 4e^{2x}$ | 1 |
| (iii) $y = 4 \log_e x$ | 1 |
- (d) Find $\int e^{3x} dx$. 1
- (e) Let $\log_a 2 = x$ and $\log_a 5 = y$. Find an expression for $\log_a 50$ in terms of x and y . 2
- (f) Write down the period of $y = 5 \tan 3x$. 1

QUESTION TWO (Start a new writing booklet)

- (a) A point moves so that its distance from the line $y = -2$ is equal to its distance from the point $(0, 6)$. The locus is a parabola of the form $(x - h)^2 = 4a(y - k)$.

- (i) Write down the coordinates of the vertex of this parabola.
 (ii) Write down the focal length.
 (iii) Write down the equation of this parabola.

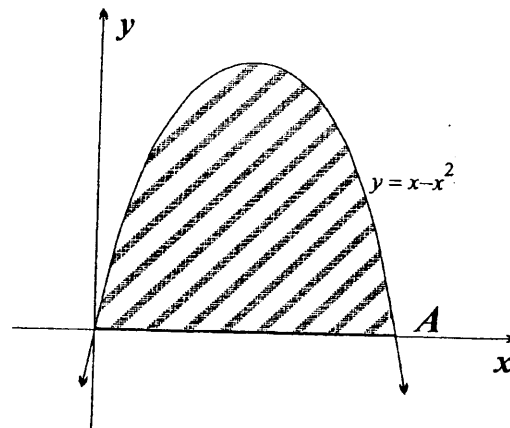
Marks

1

1

1

(b)



The diagram above shows the graph of the function $y = x - x^2$. It crosses the x -axis at A and at the origin.

- (i) Show that the x -coordinate of the point A is $x = 1$.
 (ii) Find the area of the shaded region contained by the curve $y = x - x^2$ and the x -axis.
 (iii) Find the volume obtained when this region is rotated about the x -axis. Leave your answer in terms of π .
- (c) The table below shows three values of the function $f(x)$.

x	2	4	6
$f(x)$	8	15	20

Use these three function values to approximate the value of $\int_2^6 f(x) dx$ by the trapezoidal rule.

QUESTION THREE (Start a new writing booklet)

(a) Differentiate:

(i) $f(x) = \cos(2x + 1)$

Marks

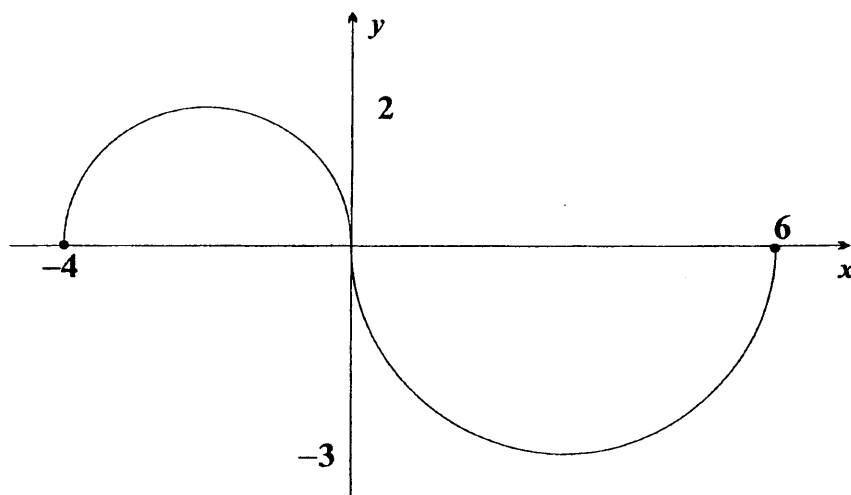
2

(ii) $f(x) = \frac{e^x}{\sin x}$

3

(b)

2



The graph of the function $f(x)$ is shown in the diagram above. It consists of two semicircles.

Evaluate $\int_{-4}^6 f(x) dx$.

(c) (i) Express 20° in radians as a multiple of π .

1

(ii) A circle has a radius of 5 cm. Find the length of the arc that subtends an angle of 20° at the centre.

1

(d) (i) If $f(x) = 27x - x^3$, find $f'(x)$.

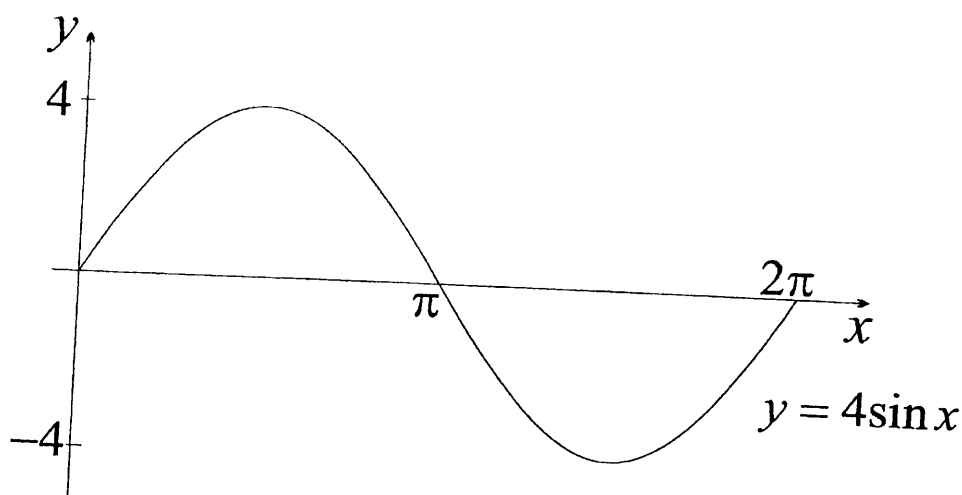
1

(ii) For what values of x is the function $f(x) = 27x - x^3$ increasing?

2

QUESTION FOUR (Start a new writing booklet)

(a)



Above is a diagram of the function $y = 4 \sin x$, for $0 \leq x \leq 2\pi$.

- (i) Copy this sketch into your writing booklet and on the same diagram sketch the line $y = -2$. How many solutions does the equation $4 \sin x = -2$ have for the domain $0 \leq x \leq 2\pi$?

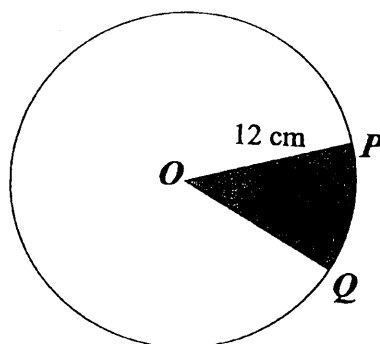
Marks

1

- (ii) Solve $4 \sin x = -2$, for $0 \leq x \leq 2\pi$.

3

(b)



The above circle has centre O and radius 12 cm. The area of the shaded sector is $18\pi \text{ cm}^2$. Let $\theta = \angle POQ$.

- (i) Show that $\theta = \frac{\pi}{4}$.

1

- (ii) Find the exact area of the minor segment cut off by the chord PQ.

2

- (c) (i) Write down the period and amplitude of $y = 3 \cos 2x$.

2

- (ii) Sketch $y = 3 \cos 2x$, for $-\pi \leq x \leq \pi$. Show clearly the intercepts with the axes.

3

QUESTION FIVE (Start a new writing booklet)

(a) Consider the function $f(x) = \sqrt{x+2}$.

(i) Show that $f'(x) = \frac{1}{2\sqrt{x+2}}$.

Marks

1

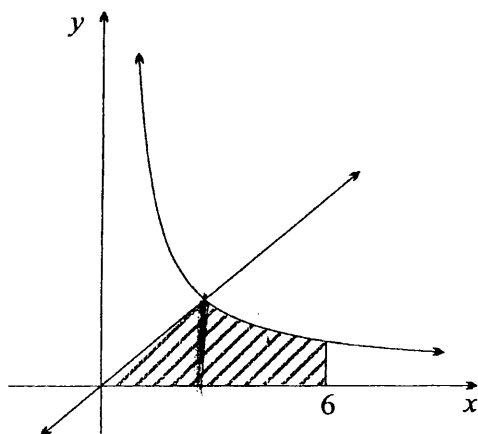
(ii) Show that the gradient of this function at the point $(7, 3)$ is $\frac{1}{6}$.

1

(iii) Find the equation of the normal to the curve $f(x) = \sqrt{x+2}$ at the point $(7, 3)$.

2

(b)



Above is a sketch of the functions $y = x$ and $y = \frac{9}{x}$.

The region between the functions and the x -axis for $0 \leq x \leq 6$ is shaded.

(i) Show that the point of intersection of these functions in the first quadrant is the point $(3, 3)$.

1

(ii) Find the exact area of the shaded region.

4

(c) By completing squares, find the centre and radius of the circle

3

$$x^2 - 6x + y^2 + 8y = 0.$$

QUESTION SIX (Start a new writing booklet)

(a) Solve $\log_e(7x - 12) = 2 \log_e x$.

Marks

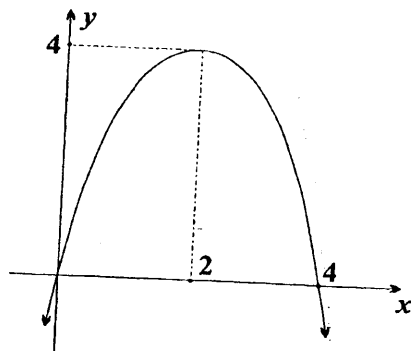
3

(b) If $f(x) = 4 \sin x + \cos x + 1$, find the exact value of $f'(x)$ when $x = \frac{\pi}{4}$.

3

(c)

2



The diagram above shows the graph of the curve $y = f'(x)$. For what value of x does $f(x)$ have a maximum turning point? Justify your answer.

(d) (i) Find $f(x)$, given that $f'(x) = \frac{2x - 5}{x^2 - 5x + 6}$ and $f(1) = 2 + \log_e 2$.

2

(ii) Hence show that $f(-1) = 2 + 2 \log_e 2 + \log_e 3$.

2

QUESTION SEVEN (Start a new writing booklet)

(a) Find the minimum value of $y = 1 + \sqrt{3} \sin x$.

Marks

1

(b) Consider the curve given by

$$y = x^3 - 6x^2 + 9x + 4.$$

(i) Show that $\frac{dy}{dx} = 3(x - 3)(x - 1)$.

1

(ii) Find the coordinates of the two stationary points.

2

(iii) Determine the nature of the stationary points.

2

(iv) Find the coordinates of the point of inflection. (Remember to justify that it is a point of inflection.)

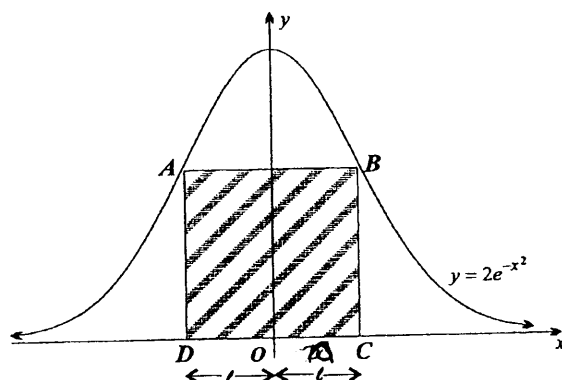
2

(v) Using half a page, sketch this function, showing clearly all important features (You need not find any x -intercepts.)

4

QUESTION EIGHT (Start a new writing booklet)

(a)



The diagram above shows the curve $y = 2e^{-x^2}$. Let D and C lie on the x -axis with $OC = OD = \ell$. Complete the rectangle $ABCD$ as shown.

- Explain why the length of BC is $2e^{-\ell^2}$.
- Show that the area of the rectangle is $A = 4\ell e^{-\ell^2}$.
- Find the value of ℓ for which $ABCD$ has maximum area.

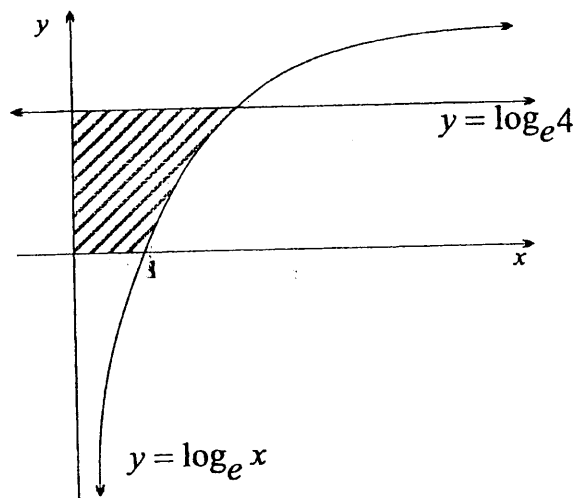
Marks

1

1

3

(b)



The diagram above shows the curve $y = \log_e x$ and the line $y = \log_e 4$.

- Solve the equation $\log_e x = \log_e 4$. Hence write down the coordinates of the point of intersection of the two functions.
- Show that the area of the shaded region is 3 square units.
- Find the exact value of $\int_1^4 \log_e x \, dx$.

1

2

1

- (c) (i) If $y = \frac{\sin x}{x}$, find $\frac{dy}{dx}$.

1

- (ii) Show that $y = \frac{\sin x}{x}$ is a solution of $\frac{dy}{dx} + \frac{y}{x} = \frac{\cos x}{x}$.

2

Solutions. 2U 2003 HY.

Q1. (a) (i) 7.389 ✓ (penalise for 3 d.pl. here only).
(ii) 2.639 ✓
(iii) 0.997 ✓

(b) $-\frac{\sqrt{3}}{2}$ ✓✓

(c) (i) $-2x^{-2}$ ✓
(ii) $8e^{2x}$ ✓
(iii) $\frac{4}{x}$ ✓

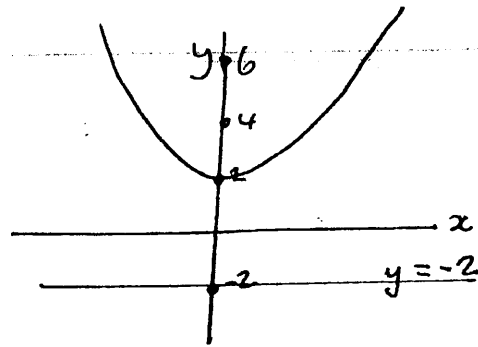
(d) $\int e^{3x} dx = \frac{1}{3}e^{3x} + c$ ✓ (don't worry about c)

(e) $\log_2 50 = \log_2 (2 \times 5 \times 5)$
 $= \log_2 2 + \log_2 5 + \log_2 5$ ✓
 $= x + 2y$ ✓

(f) $\frac{\pi}{3}$ ✓

Q2.

(a)



- (i) $(0, 2)$ ✓
 (ii) 4 ✓
 (iii) $x^2 = 16(y - 2)$ ✓

(b) (i) A is on the x axis where $y = 0$

So solve $x - x^2 = 0$

$$x(1 - x) = 0$$

$$x = 0, 1$$

So $x = 1$ is one of the solutions ✓

(ii) Area = $\int_0^1 (x - x^2) dx$ ✓

$$= \left[\frac{x^2}{2} - \frac{x^3}{3} \right]_0^1$$

$$= \left(\frac{1}{2} - \frac{1}{3} \right) - (0)$$

$$= \frac{1}{6} \text{ u}^2$$
 ✓

(iii) Volume = $\pi \int_0^1 y^2 dx$

$$= \pi \int_0^1 (x - x^2)^2 dx$$
 ✓
$$= \pi \int_0^1 (x^2 - 2x^3 + x^4) dx$$

$$= \pi \left[\frac{x^3}{3} - \frac{2x^4}{4} + \frac{x^5}{5} \right]_0^1$$
 ✓
$$= \pi \left[\left(\frac{1}{3} - \frac{1}{2} + \frac{1}{5} \right) - (0) \right]$$

$$= \frac{\pi}{30} \text{ u}^3$$
 ✓

$$\begin{aligned}
 (a) \quad \int_2^6 f(x) dx &\approx \frac{h}{2} [f(2) + f(4)] + \frac{h}{2} [f(4) + f(6)] \\
 &= \frac{2}{2} [8 + 15 + 15 + 20] \quad \checkmark \text{ for correct substitution into correct formula} \\
 &= 58 \quad \checkmark
 \end{aligned}$$

Q3.

(a) i) $f(x) = \cos(2x+1)$

$$f'(x) = -2\sin(2x+1) \quad \checkmark \quad \checkmark$$

ii) $f(x) = \frac{e^x}{\sin x}$

$$f'(x) = \frac{e^x \sin x - e^x \cos x}{\sin^2 x} \quad \checkmark \quad \checkmark \quad \checkmark$$

$$= \frac{e^x (\sin x - \cos x)}{\sin^2 x}$$

(b) $\int_{-4}^6 f(x) dx = \frac{\pi}{2} 2^2 - \frac{\pi}{2} 3^2 \quad \checkmark$

$$= \frac{\pi}{2} (4-9)$$

$$= -\frac{5\pi}{2} \quad \checkmark$$

(c) i) $180^\circ = \pi$ radians

$$20^\circ = \frac{\pi}{9} \text{ radians} \quad \checkmark$$

ii) $\ell = r\theta$

$$= 5 \times \frac{\pi}{9} \text{ cm}$$

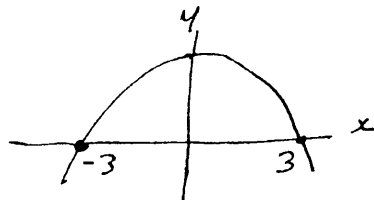
$$= \frac{5\pi}{9} \text{ cm} \quad \checkmark$$

(d) i) $f(x) = 27x - x^3$

$$f'(x) = 27 - 3x^2 \quad \checkmark$$

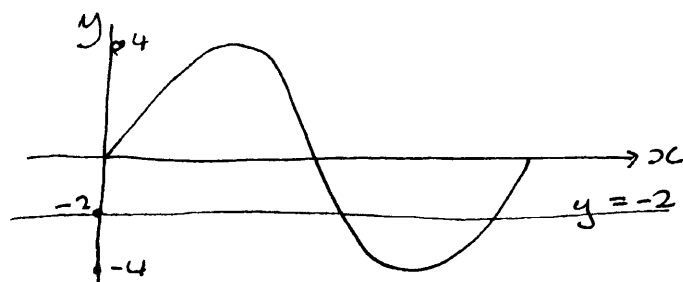
$$= 3(3-x)(3+x)$$

ii) Solve $3(3-x)(3+x) > 0 \quad \checkmark$



$$-3 < x < 3 \quad \checkmark$$

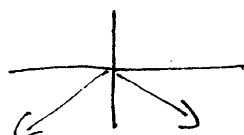
24.
a)



(i) 2 solutions

✓ (need leave $y=2$
and 2 soln
for this mk)

(ii) $4 \sin x = -2$
 $\sin x = -\frac{1}{2}$



reference angle is $\frac{\pi}{6}$

$$x = \pi + \frac{\pi}{6}, 2\pi - \frac{\pi}{6}$$

$$= \frac{7\pi}{6}, \frac{11\pi}{6}$$

ch1 (i) Area = $\frac{1}{2} r^2 \theta$
 $18\pi = \frac{1}{2} \times 12^2 \times \theta$
 $\theta = \frac{18\pi \times 2}{144}$
 $= \frac{\pi}{4}$

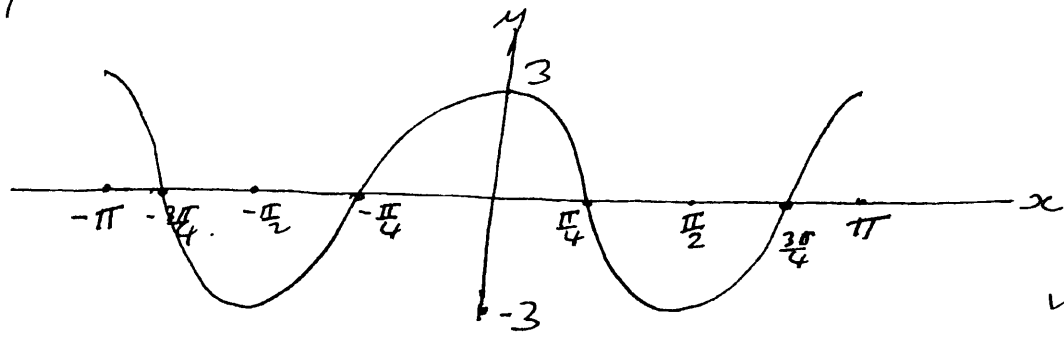
(ii) Area $\triangle OPQ = \frac{1}{2} \times 12 \times 12 \times \sin \frac{\pi}{4}$ ✓
 $= \frac{72}{\sqrt{2}} \text{ cm}^2$

Area of segment = $18\pi - \frac{72}{\sqrt{2}} \text{ cm}^2$ ✓

or/ Area segment = $\frac{1}{2} r^2 (\theta - \sin \theta)$ ✓
 $= 72 \left(\frac{\pi}{4} - \frac{1}{\sqrt{2}} \right) \text{ cm}^2$ ✓

(c) (i) period is $\frac{2\pi}{2} = \pi$ ✓
amplitude is 3. ✓

(ii)



5 (a) (i) $f(x) = (x+2)^{\frac{1}{2}}$
 $f'(x) = \frac{1}{2}(x+2)^{-\frac{1}{2}}$ ✓
 $= \frac{1}{2\sqrt{x+2}}$

(ii) $f'(7) = \frac{1}{2\sqrt{7+2}}$ is gradient ✓
 $= \frac{1}{6}$

(iii) gradient of normal is -6 ✓
 equation of normal is
 $y-3 = -6(x-7)$
 $y-3 = -6x+42$ ✓
 $y+6x-45=0$

4b-1(i) (3,3) clearly satisfies $y=x$.
 Consider $y = \frac{9}{x}$, if $x=3$ then $y = \frac{9}{3} = 3$

So (3,3) lies on both curves, so it is a point of intersection. ✓

(ii) Area = area triangle + area under curve ✓
 $= \frac{1}{2} \times 3 \times 3 + \int_3^6 \frac{9}{x} dx$ ✓
 $= \frac{9}{2} + [9 \log_e x]_3^6$ ✓
 $= \frac{9}{2} + 9 [\log 6 - \log 3]$
 $= \frac{9}{2} + 9 \log 2$ ✓ ✓

(c) $x^2 - 6x + 9 + y^2 + 8y + 16 = 9 + 16$ ✓
 $(x-3)^2 + (y+4)^2 = 25$
 centre is (3, -4), radius is 5 ✓✓

Q6. (a) $\log_e (7x-12) = 2 \log_e x$
 $\log_e (7x-12) = \log_e x^2$ ✓
 $7x-12 = x^2$
 $x^2 - 7x + 12 = 0$ ✓
 $(x-3)(x-4) = 0$
 $x = 3, 4.$ ✓

(b) $f(x) = 4 \sin x + \cos x + 1$
 $f'(x) = 4 \cos x - \sin x$ ✓
 $f'(\pi/4) = 4 \cos \pi/4 - \sin \pi/4$ ✓
 $= \frac{4}{\sqrt{2}} - \frac{1}{\sqrt{2}}$
 $= \frac{3}{\sqrt{2}}$ ✓

(c) There is a maximum turning point at $x=4$.
 The gradient, $f'(x)$, changes from positive to negative at $x=4$. ✓

x	3	4	5
$f'(x)$	+	0	-

(d) (i) $f(x) = \int \frac{2x-5}{x^2-5x+6} dx$
 $= \log_e (x^2-5x+6) + c$ ✓
 $f(1) = \log_e (1-5+6) + c = 2 + \log_e 2$
 $\log_e 2 + c = \log_e 2 + 2$
 $c = 2$
 so $f(x) = \log_e (x^2-5x+6) + 2$ ✓

(ii) $f(-1) = \log_e (1+5+6) + 2$ ✓
 $= \log_e 12 + 2$
 $= \log(2^2 \times 3) + 2$ ✓
 $= 2 \log 2 + \log 3 + 2$

Q7. (a) Minimum value is when $\sin x = -1$.
Min. $y = 1 - \sqrt{3}$. ✓

(b) (i) $y = x^3 - 6x^2 + 9x + 4$
 $\frac{dy}{dx} = 3x^2 - 12x + 9$
 $= 3(x^2 - 4x + 3)$
 $= 3(x-1)(x-3)$ ✓

(ii) At stationary point $\frac{dy}{dx} = 0$.

$$3(x-1)(x-3) = 0$$

$$x = 1, 3$$

$$x=1, y=8 \text{ and } x=3, y=4$$

Stationary points are $(1, 8)$ and $(3, 4)$ ✓

(iii) $\frac{d^2y}{dx^2} = 6x - 12$

when $x=1$, $\frac{d^2y}{dx^2} = 6 - 12 = -6$, negative ✓
so $(1, 8)$ is a maximum turning point.

when $x=3$, $\frac{d^2y}{dx^2} = 18 - 12 = 6$, positive ✓
so $(3, 4)$ is a minimum turning point.

(iv) At a possible point of inflection, $\frac{d^2y}{dx^2} = 0$

$$6x - 12 = 0$$

$$x = 2$$

$$y = 6$$

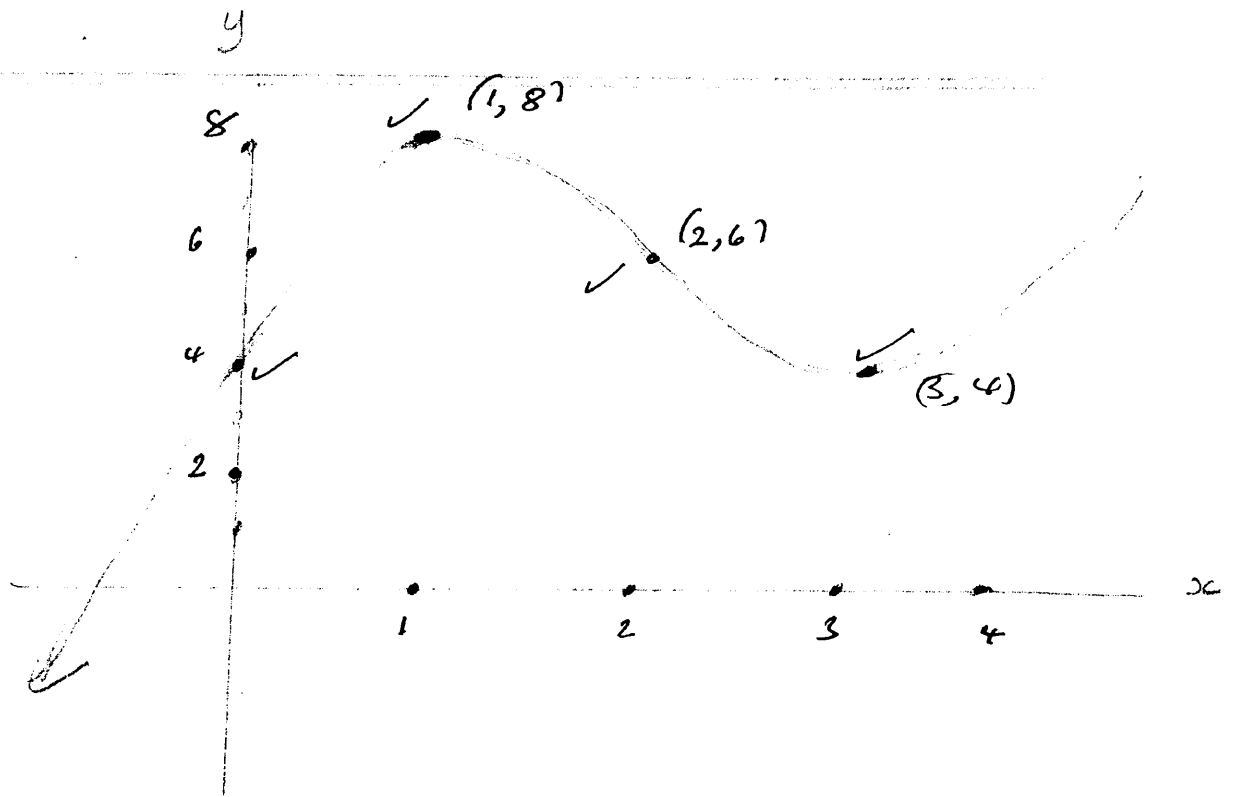
$(2, 6)$ may be a pt of inflection ✓

x	1	2	3
$\frac{d^2y}{dx^2}$	-6 ∩	0 -	6 ∪

ie concavity changes
at $(2, 6)$ ✓

So $(2, 6)$ is a point of inflection.

(07)



Q8.

(i) at C , $x=l$
 So B is $(l, 2e^{-l^2})$
 and BC is $2e^{-l^2}$ units

(ii) Area = $OC \times BC$
 $= 2l \times 2e^{-l^2}$
 $= 4le^{-l^2}$

(iii) $\frac{dA}{dl} = 4l \cdot (-2le^{-l^2}) + 4e^{-l^2}$
 $= 4e^{-l^2}(1 - 2l^2)$

now, $4e^{-l^2}(1 - 2l^2) = 0$ at a stationary point
 $l^2 = \frac{1}{2}$
 $l = \pm \frac{1}{\sqrt{2}}$

choose $l = \frac{1}{\sqrt{2}}$ as l is a length.

To check for maximum

l	$\frac{1}{2}$	$\frac{1}{\sqrt{2}}$	1
$\frac{dA}{dl}$	$4e^{-\frac{1}{4}} \times \frac{1}{2}$	0	$-4e^{-1}$
	+	-	-

so we have maximum A
 when $l = \frac{1}{\sqrt{2}}$.

(i) $\log x = \log 4$
 $x = 4$

Pt of intersection is $(4, \log 4)$

(ii) $y = \log_e x$
 $e^y = x$
 $A = \int_0^{\log 4} e^y dy$
 $= [e^y]_0^{\log 4}$
 $= [4 - e^0]$
 $= 4 - 1 = 3$

$$\text{Giii)} \quad \int_1^4 \log_e x \, dx = \text{area of rectangle} - \text{shaded area} \quad \checkmark$$

$$= 4 \log 4 - 3$$

$$\text{(c) (i)} \quad y = \frac{\sin x}{x}$$

$$\frac{dy}{dx} = \frac{x \cos x - \sin x}{x^2} \quad \checkmark$$

$$= \frac{\cos x}{x} - \frac{\sin x}{x^2}$$

$$\text{(ii)} \quad \text{LHS} = \frac{dy}{dx} + \frac{y}{x}$$

$$= \frac{\cos x}{x} - \frac{\sin x}{x^2} + \frac{\sin x}{x} \cdot \frac{1}{x} \quad \checkmark$$

$$= \frac{\cos x}{x} \quad \checkmark$$

$$= \text{RHS}$$